

Chapter #10:

Software Liability: Liability & Practice

Introduction:

Software is used in many critical areas like manufacturing, medicine, transportation and even military applications.

Industry:

Manufacturing systems, robots etc.

Medicine:

Intensive care monitoring

Transport:

Railway signaling systems, aircraft, spaceshuttle

Military & defense applications.

⇒ There should be legal responsibility of those who create and use software

⇒ if a software system causes harm the creators and users can held accountable.

⇒ These are some rules and regulations to ensure software safety. These regulations help to prevent accidents and protect people.

Regulatory Issues

⇒ Regulatory issues refer to the rules, laws and standards that govern the development, distribution and use of software. These regulations are put in place to ensure safety, security and ethical practices in the software industry.

1. Standards:

0 These are rules and guidelines (standards) that software needs to meet to make sure it's safe and works as it should. This is like a check list to ensure quality.

2. Certification:

0 Sometimes, software needs to be officially approved (certified) to show that it meets these standards. It is like getting a stamp of approval.

3. Licensing:

Some times, might need a special permit (license) to use certain software, especially if it's powerful or has the potential to cause harm.

4. Professional codes of practice:

- Self Regulations: Professional organizations and associations create their own guidelines (Codes of practice) for their members to follow. This is like a self-imposed set of rules to ensure good behavior and quality work.

5. Regulation by law:

o Government involvement: Governments also have laws and regulations that software needs to comply with. These laws can be directly about software or can indirectly affect it through other regulations like safety standards and licensing requirements.

o Fears of legal Trouble:

Companies and developers are motivated to follow these regulations because they fear legal consequences if their software causes harm or does not meet safety standards.

Legal liability

⇒ System designers and software-engineers may have legal responsibility.

Product Liability & Consumer protection Act 1987:

o Product liability is the area of law in which manufacturers, distributors, suppliers and retailers are held responsible for any injuries product cause.

o Consumer protection Act (CPA) can hold software companies accountable for a defective product or if any of its component parts are defective.

Negligence:

o Carelessness can lead to problems: if a software developer or company is careless in creating or designing a software, and this leads to problems or injuries, they can be held responsible.

o Responsibility for mistakes:

if the software fails because of a mistake made during its creation.

the company or developer can be liable for any damage caused.

Competence, Training & Experience:

• Assumption of Competence:

⇒ Competence means "knowledge and the ability to apply that knowledge".
it is assumed that people working on safety-critical software are qualified and know what they are doing.

• Training & Experience:

Having the right training and experience is important for being a good software developer.

• Knowledge & Skills:

Software developers need to have the knowledge and skills to do their job properly.

Factors affecting System Safety:

- Hazard analysis
- Requirement & Specification
- System Reliability & Safety
- Design
- Documentation
- Testing & debugging
- Safety integrity analysis & risk assessment